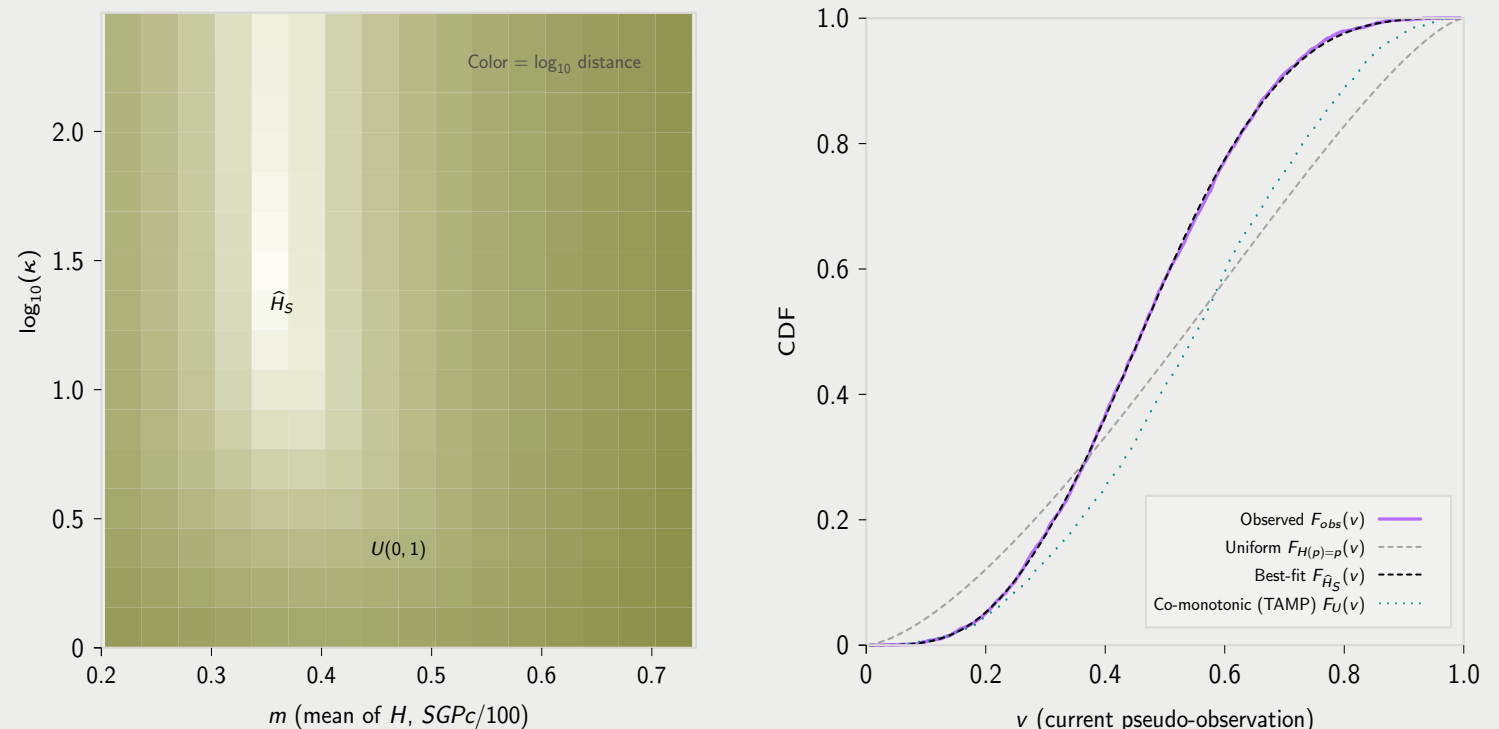


B1 & B2. Estimate subgroup growth regime $\hat{H}_S$ (copula + regime $\Rightarrow$ induced marginal)

Choose $H \in \mathcal{H}_{\text{Beta}}$ to minimize $W_1(F_{V,S}^{\text{obs}}, F_H)$, where $F_H(v) = \mathbb{E}_U[H(F_0(v \mid U))]$



The mathematical bridge connecting the observed, unlinked marginals (Panel A) and the recovered latent SGPC regime (Panel C) has two ingredients:

1. A baseline copula C_0 providing a dependence template from prior reference-quantiles U to current reference-quantiles V .²
2. A stochastic growth regime H that governs how subgroup S allocates conditional percentiles within that template.^{3, 4}

Together, C_0 and H define a *forward map* from a candidate regime to a *predicted* current-grade marginal CDF. Panel B2 visualizes this map via the induced $F_H(v)$, and Panel B1 *inverts* it by choosing the regime whose F_H best matches the observed current marginal $F_{V,S}^{\text{obs}}$.

Let C_0 be a baseline copula. C_0 induces the conditional CDF (transition kernel)

$$F_0(v \mid u) = \Pr(V \leq v \mid U = u) = \frac{\partial}{\partial u} C_0(u, v).$$

Its inverse in v (the conditional quantile kernel) is

$$Q_0(p \mid u) = F_0^{-1}(p \mid u).$$

Intuitively, for each starting quantile u , the kernel $F_0(\cdot \mid u)$ describes the distribution of possible ending quantiles V under the baseline dependence structure C_0 .

Let $P \in [0, 1]$ be a random variable (a latent SGPC on the 0–1 scale) and let p denote a realization in $[0, 1]$. A *growth regime* H is the distribution of P , represented by its CDF

$$H(p) := \Pr(P \leq p), \quad p \in [0, 1].$$

Given $U = u$ and a draw $P = p$, the model generates an outcome percentile by:⁵

$$V = Q_0(P \mid U = u).$$

The uniform regime $H(p) = p$ (i.e., $P \sim \text{Uniform}(0, 1)$) corresponds to occupying the baseline kernel with no tilt. Departures from $H(p) = p$ tilt the growth regime mass toward lower or higher conditional percentiles relative to the baseline regime.

Although we do not observe linked pairs (u, v) in TIMSS- or NAEP-style settings, the model implies a forward map from the observed prior marginal F_U to the predicted current marginal F_H . For any fixed u ,

$$\begin{aligned} \Pr(V \leq v \mid U = u) &= \Pr(Q_0(P \mid u) \leq v) \\ &= \Pr(P \leq F_0(v \mid u)) \\ &= H(F_0(v \mid u)). \end{aligned}$$

Therefore, by the law of total probability, the subgroup’s predicted current-grade marginal CDF is

$$\begin{aligned} F_H(v) &= \Pr(V \leq v) \\ &= \mathbb{E}_U[H(F_0(v \mid U))] \\ &= \int_0^1 H(F_0(v \mid u)) dF_U(u) \\ &\approx \frac{\sum_{i=1}^n w_i H(F_0(v \mid u_i))}{\sum_{i=1}^n w_i}, \end{aligned}$$

where $\{u_i, w_i\}$ are the observed prior pseudo-observations and survey weights for the subgroup (i.e., a weighted empirical approximation to F_U).

We estimate the subgroup’s growth regime by searching over an admissible class $\mathcal{H}$ of candidate CDFs on $[0, 1]$. Panel B1 shows the heat map of $\mathcal{H}$ over the *Beta regime family*

$$\mathcal{H}_{\text{Beta}} = \{H_{m,\kappa}(\cdot) : P \sim \text{Beta}(\kappa m, \kappa(1 - m)), \ 0 < m < 1, \ \kappa > 0\},$$

where $m = \mathbb{E}[P]$ controls the regime center (mean SGPC on the 0–1 scale) and κ controls concentration (dispersion of latent SGPC).⁶

For each candidate $H \in \mathcal{H}$, Panel B2 provides the induced *predicted* current-grade marginal

$$F_H(v) = \mathbb{E}_U[H(F_0(v \mid U))].$$

The growth regime estimate for subgroup S is defined by the minimum-distance problem

$$\hat{H}_S = \underset{H \in \mathcal{H}_{\text{Beta}}}{\operatorname{argmin}} W_1(F_{V,S}^{\text{obs}}, F_H),$$

where

$$W_1(F, G) = \int_0^1 |F(v) - G(v)| dv$$

is the Wasserstein-1 (earth mover’s) distance between the observed and predicted current-grade marginal CDFs.⁷

The heat map displays $\log_{10}$ Wasserstein-1 distance over $(m, \log_{10} \kappa)$: lighter regions indicate candidate regimes whose induced F_H more closely matches $F_{V,S}^{\text{obs}}$. The marked optimum corresponds to $\hat{H}_S$ (equivalently, $(\hat{m}, \hat{\kappa})$). For reference, the uniform regime $H(p) = p$ occurs at $m = 0.5$ and $\kappa = 2$ (i.e., $\text{Beta}(1, 1)$), indicated by the $U(0, 1)$ marker.

The baseline copula, C_0 , combined with the best fit growth regime estimate, $\hat{H}_S$, induces the Wasserstein-1 minimized CDF, $F_{\hat{H}_S}(v)$, and its density, $f_{\hat{H}_S}(v)$, which are shown in panel B2 and A, respectively, with a dashed-black line.

² The baseline copula used for inference is selected from canonical STEP 1 sensitivity analyses over 966 longitudinal conditions across four datasets: the t -copula is best by AIC in 63.6% of conditions (Frank: 30.7%). In the cross-stratified canonical table below (content area $\times$ year span), the recommended family is t in every cell; entries report (p, df) medians.

Content area	1-year	2-year	3-year	4-year
MATHEMATICS	(0.85, 25.26)	(0.81, 27.40)	(0.78, 28.38)	(0.76, 27.34)
READING	(0.84, 28.33)	(0.80, 29.34)	(0.77, 26.63)	(0.76, 28.22)
WRITING	(0.85, 23.72)	(0.82, 27.58)	(0.80, 27.22)	(0.77, 23.86)
ELA	(0.84, 41.95)	(0.79, 46.41)	(0.76, 51.57)	(0.72, 42.04)

³ Other regime families considered include the truncated exponential family (the maximum-entropy distribution on a finite interval with a fixed mean) and the truncated uniform family. If acceptable accuracy and precision are already achieved under the Beta family, additional growth-regime optimization and fine-tuning should only improve the recovered fit.

⁴ Here, U denotes the subgroup’s prior-grade reference quantile (pseudo-observation), and C_0 is the canonical baseline copula selected from STEP 1 by content area and year span. The fitted growth regime $\hat{H}_S$ (distribution of latent $P = \text{SGPC}/100$) and C_0 define a forward map

$$(U, C_0, \hat{H}_S) \mapsto F_{\hat{H}_S}(v) = \mathbb{E}_U[\hat{H}_S(F_0(v \mid U))],$$

with corresponding induced density $f_{\hat{H}_S}(v)$. The latent SGPC distribution itself is given by $\hat{H}_S$ (density $\hat{h}_S(p) = d\hat{H}_S/dp$), which is estimated by matching $F_{\hat{H}_S}$ to the observed current marginal.

⁵ We assume $P \perp U$. That is, p is drawn independently of starting point within the subgroup; meaning there is no correlation between the starting point u and the growth regime. This assumption is investigated more deeply in the full paper.

⁶ The Beta family, $\text{Beta}(\alpha, \beta)$, is reparameterized via $\alpha = \kappa m$ and $\beta = \kappa(1 - m)$, so that $m = \mathbb{E}[P] = \alpha/(\alpha + \beta)$ (the regime center, mean SGPC on the 0–1 scale) and $\kappa = \alpha + \beta$ (the concentration parameter).

⁷ Alternative minimum-distance criteria (e.g., Cramér–von Mises, $\int_0^1 \{F(v) - G(v)\}^2 dv$) are also valid for regime fitting. If acceptable accuracy and precision are already achieved under Wasserstein-1, then additional growth-regime optimization and fine-tuning should only improve the recovered fit.